Lecture 10. Monte Carlo Integration

Naive Monte Carlo, importance sampling, antithetic variables, control variables

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10.1 Naive Monte Carlo

To compute $\mu = \mathbb{E}f(X) = \int f(x)p(x)\,dx$, draw $X_1, \dots, X_n$ i.i.d. from p and use

$$\hat{\mu}_n = \frac{1}{n}\sum_{i=1}^{n} f(X_i).$$

It is unbiased, $\hat{\mu}_n \to \mu$ a.s. when $\mathbb{E}|f(X)| < \infty$, and $\sqrt{n}(\hat{\mu}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$ when $\sigma^2 = \operatorname{Var} f(X) < \infty$; with the sample variance $\hat{\sigma}^2$, $\hat{\mu}_n \pm 1.96\,\hat{\sigma}/\sqrt{n}$ is an approximate 95% interval.

The error is $O_p(n^{-1/2})$ *in every dimension*, while grid quadrature suffers the curse of dimensionality: quadrature (Lecture 8) wins in one or two dimensions, Monte Carlo in high dimensions. The rate cannot be improved much, but σ^2 can: **variance reduction**.

Example 10.1

- $\int_0^\infty \sqrt{x}e^{-x}dx = \Gamma(3/2) = \sqrt{\pi}/2 = \mathbb{E}\sqrt{X}$, $X \sim \text{Exp}(1)$, with finite variance.
- $\int_0^1 x^{-1/2}dx = 2 = \mathbb{E}U^{-1/2}$, but $\mathbb{E}U^{-1} = \infty$: consistent with infinite variance, so the CLT interval is meaningless and running averages jump. Substituting $x = y^2$ gives $\int_0^1 2\,dy$, estimated exactly: flattening the integrand is the cheapest variance reduction.
- Buffon's needle: a needle of length L on lines spaced $D \geq L$ crosses one with probability $2L/(\pi D)$. With centre distance $Y \sim \text{Unif}(0, D/2)$ and angle $\Theta \sim \text{Unif}(0, \pi/2)$ independent, it crosses iff $Y \leq \frac{L}{2}\sin\Theta$, with probability $\frac{2}{D} \cdot \frac{2}{\pi}\int_0^{\pi/2} \frac{L}{2}\sin\theta\,d\theta = \frac{2L}{\pi D}$. With k crossings in n throws, $\hat{p} = k/n$ is unbiased for p, and $\hat{\pi} = 2Ln/(Dk)$; it is biased ($E[1/\hat{p}] > 1/p$ by Jensen), indeed $E\hat{\pi} = \infty$ as $P(k = 0) > 0$. The delta method with $g(p) = 2L/(Dp)$, $g'(p)^2 = \pi^2/p^2$, gives

  $$\operatorname{Var}\hat{\pi} \approx g'(p)^2 \frac{p(1-p)}{n} = \frac{\pi^2(1-p)}{np},$$

  decreasing in p; $L = D$ gives $\operatorname{Var}\hat{\pi} \approx 5.63/n$, so three digits need tens of millions of throws.

10.2 Importance sampling

For a density $h > 0$ wherever $fp \neq 0$,

$$\mu = \int f(x)\frac{p(x)}{h(x)}h(x)\,dx = \mathbb{E}_h\Big[f(Y)\frac{p(Y)}{h(Y)}\Big], \qquad \hat{\mu}_h = \frac{1}{n}\sum_{i=1}^{n} f(Y_i)w(Y_i), \quad Y_i \overset{\text{iid}}{\sim} h,$$

with **importance weights** $w = p/h$: unbiased, variance $\frac{1}{n}\{\int f^2p^2/h - \mu^2\}$. Useful when p is hard to sample or rarely visits where f is large.

Proposition 10.1

The variance of $\hat{\mu}_h$ is minimized by $h^\star \propto |f|\,p$. If $f \geq 0$, then $h^\star = fp/\mu$ and the variance is zero.

Proof. By Cauchy–Schwarz, for admissible h,

$$\Big(\int |f|p\Big)^2 = \Big(\int \frac{|f|p}{\sqrt{h}}\sqrt{h}\Big)^2 \leq \int \frac{f^2p^2}{h}\int h = \int \frac{f^2p^2}{h},$$

with equality iff $h \propto |f|p$. If $f \geq 0$, $fw \equiv \mu$ under $h^\star$. ■

$h^\star$ needs μ, but guides the choice: h roughly proportional to $|f|p$, with tails at least as heavy as fp, or the weights explode and the variance may be infinite. Diagnose with the spread of the weights.

Example 10.2 (Normal tail probability)

Estimate $\pi(c) = P(Z > c) = \Phi(-c)$ for large c. The naive estimator has relative error $\sqrt{(1-\pi(c))/(n\pi(c))}$, exploding in c. Sample $Y_i \sim N(c,1)$ instead; the weight is $\phi(y)/\phi(y-c) = e^{-cy+c^2/2}$, so

$$\hat{\pi}_{\text{imp}}(c) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{Y_i > c\} e^{-cY_i + c^2/2}.$$

Its second moment per draw is

$$e^{c^2}\int_c^\infty e^{-2cy}\phi(y-c)\,dy = e^{c^2}\int_c^\infty \phi(y+c)\,dy = e^{c^2}\Phi(-2c),$$

as $-2cy - \frac{1}{2}(y-c)^2 = -\frac{1}{2}(y+c)^2$; so $\operatorname{Var}\hat{\pi}_{\text{imp}}(c) = \frac{1}{n}\{e^{c^2}\Phi(-2c) - \pi(c)^2\}$. By Mills' ratio $\Phi(-t) \approx \phi(t)/t$, the squared relative error is about $c\sqrt{\pi/2}/n$ against $\sqrt{2\pi}\, ce^{c^2/2}/n$ naively: the relative error grows like $c^{1/2}$ instead of $c^{1/2}e^{c^2/4}$.

10.3 Antithetic variates

If $X \overset{d}{=} X'$ are negatively correlated,

$$\operatorname{Var}\left[\frac{f(X)+f(X')}{2}\right] = \frac{\operatorname{Var} f(X)}{2}(1+\rho), \qquad \rho = \operatorname{Corr}(f(X), f(X')),$$

beating two independent draws when $\rho < 0$. Natural pairs: $(U, 1-U)$ for uniforms, $(X, 2m-X)$ for distributions symmetric about m.

Proposition 10.2

If f and g are both nondecreasing (or both nonincreasing), then $\operatorname{Cov}(f(X), g(X)) \geq 0$ whenever it exists. Hence $\operatorname{Cov}(h(U), h(1-U)) \leq 0$ for monotone h.

Proof. With Y an independent copy of X, $\{f(X)-f(Y)\}\{g(X)-g(Y)\} \geq 0$, with expectation

$$2\mathbb{E}f(X)g(X) - 2\mathbb{E}f(X)\mathbb{E}g(X) = 2\operatorname{Cov}(f(X), g(X)) \geq 0.$$

For the consequence take $f = h$, $g(u) = -h(1-u)$, monotone in the same direction. ■

Example 10.3

For $\mu = \mathbb{E}e^U = e - 1$: $\operatorname{Var} e^U = \frac{e^2-1}{2} - (e-1)^2 \approx 0.2420$ and $\operatorname{Cov}(e^U, e^{1-U}) = e - (e-1)^2 \approx -0.2342$, so $\frac{1}{2}(e^U + e^{1-U})$ has variance ≈ 0.0039 against 0.1210 for two independent draws: about 97% less.

10.4 Control variates

If $Y = g(X)$ has known mean μ_Y, $f(X) - c(Y - \mu_Y)$ is unbiased for every c, with variance

$$\operatorname{Var} f(X) + c^2 \operatorname{Var} Y - 2c\operatorname{Cov}(f(X), Y),$$

minimized at $c^\star = \mathrm{Cov}(f(X), Y)/\operatorname{Var} Y$, giving $\operatorname{Var} f(X)\,(1-\rho^2)$, $\rho = \mathrm{Corr}(f(X), Y)$. In practice $c^\star$ is the least-squares slope of $f(X_i)$ on Y_i (negligible bias for large n).

Example 10.4

For $\mathbb{E}e^U$ with $Y = U$ ($\mu_Y = \frac{1}{2}$, $\operatorname{Var} U = \frac{1}{12}$): $\mathrm{Cov}(e^U, U) = 1 - \frac{e-1}{2} \approx 0.1409$, $c^\star \approx 1.6903$, and the variance per draw drops from 0.2420 to $0.2420 - 12(0.1409)^2 \approx 0.0039$.

10.5 A comparison: the size of a test

Example 10.5

For $X_1, \ldots, X_{25}$ i.i.d. Poisson(λ), the CLT test of $H_0 : \lambda = 2$ vs $\lambda > 2$ rejects when $(\bar{X} - 2)/\sqrt{2/25} > 1.645$, i.e. $S = \sum X_i > c = \lfloor 25(2 + 1.645\sqrt{2/25}) \rfloor = 61$. As $S \sim \mathrm{Poi}(25\lambda)$, the size is $p = 1 - F_{50}(61) = 0.05568$. Estimate p by N simulated S:

- *Standard*: $\hat{p} = \frac{1}{N}\sum \mathbf{1}\{S_j > c\}$ has variance $p(1-p)/N = 0.0526/N$.
- *Antithetic*: with $S_j = F_{50}^{-1}(U_j)$, $S_j' = F_{50}^{-1}(1-U_j)$, both indicators equal 1 only if $F_{50}(c) < U_j < 1 - F_{50}(c)$, impossible as $F_{50}(c) > \frac{1}{2}$; so the covariance is $-p^2$ and N pairs have variance $p(1-2p)/(2N) = 0.0247/N$ against $0.0263/N$ for $2N$ independent draws. Only 6%: a rare-event indicator is monotone but nearly constant.
- *Importance sampling*: $S_j \sim h = \mathrm{Poi}(25\lambda')$ with $\lambda' = 2.4653$ (the threshold, so about half the draws reject), weight $w(s) = e^{-25(\lambda - \lambda')}(\lambda/\lambda')^s$; the variance per draw is $\sum_{s>c} f_{50}(s)^2/h(s) - p^2 = 0.0063$ (f_{50} the Poi(50) pmf): 8.3 times smaller.

Each gives a CLT interval $\hat{p} \pm z_{1-\alpha/2}\sqrt{\widehat{\mathrm{Var}}}$. For the power at $\lambda \in [2.2, 4]$, replace F_{50} by $F_{25\lambda}$. Once rejection is not rare ($\lambda \geq 2.5$, power ≥ 0.5) the fixed envelope $\lambda' = 2.4653$ loses its advantage, and for large λ the weights $(\lambda/\lambda')^s$ make importance sampling *worse* than standard Monte Carlo: per draw 0.299 vs 0.248 at $\lambda = 2.5$, and 17.3 vs 0.053 at $\lambda = 3$. An envelope tuned to one λ does not suit all.

10.6 Selected exercises

Motivation. How too-light tails of the sampling density ruin importance sampling, exactly, in a Gaussian case.

Exercise 10.1 (Importance densities with light tails)

Let $p = N(0,1)$ and $h = N(0, s)$ with variance $s > 0$, and $w = p/h$.
(a) Show that $\mathbb{E}_h[w(Y)^2] < \infty$ if and only if $s > \frac{1}{2}$, and compute it.
(b) Using $f \equiv 1$ (so that the estimator is the average weight, estimating 1), find the variance per draw and the best s.

Solution. [Supplement]

(a) $\mathbb{E}_h[w^2] = \int p^2/h$, and

$$\frac{p(x)^2}{h(x)} = \frac{e^{-x^2}/(2\pi)}{e^{-x^2/(2s)}/\sqrt{2\pi s}} = \sqrt{\frac{s}{2\pi}}\exp\Big\{-x^2\Big(1 - \frac{1}{2s}\Big)\Big\}.$$

For $s \le \frac{1}{2}$ the exponent is ≥ 0 and the integral diverges. For $s > \frac{1}{2}$, with $\int e^{-cx^2}dx = \sqrt{\pi/c}$, $c = 1 - \frac{1}{2s} > 0$,

$$\mathbb{E}_h[w^2] = \sqrt{\frac{s}{2\pi}}\sqrt{\frac{\pi}{1-1/(2s)}} = \frac{s}{\sqrt{2s-1}}.$$

(b) The variance per draw is $s/\sqrt{2s-1} - 1$, with derivative $(s-1)/(2s-1)^{3/2}$: decreasing on $(\frac{1}{2}, 1)$, increasing on $(1, \infty)$, zero at $s = 1$ ($h = p$). It blows up as $s \downarrow \frac{1}{2}$ and is infinite for $s \le \frac{1}{2}$, though the estimator stays unbiased and consistent: lighter tails than $|f|p$ give rare, enormous weights. Heavier tails cost moderately, e.g. $\frac{2}{\sqrt{3}} - 1 \approx 0.155$ at $s = 2$. ■